Software Requirements Specification

for

UncleSam Travel and Tours

Version <X.X>

Prepared by

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Revisions

Version	Primary Author(s)	Description of Version	Date Completed
Draft Type and Number	Full Name	Information about the revision. This table does not need to be filled in whenever a document is touched, only when the version is being upgraded.	00/00/00

1 Introduction

<TO DO: Please provide a brief introduction to your project and a brief overview of what the reader will find in this section.>

1.1 Document Purpose

UncleSam Travel & Tours is a booking webapp designed solely for our client's Japan based company. It is intended to enhance the experience of Japan travellers. Important characteristics of the system will include -- Itinerary generator with AI chatbot/integrated system, Integration of payment gateways (PayMongo), weather forecasting in real time, features for searching and filtering, currency conversion for payment. In addition to the user dashboard, we are developing a basic admin dashboard that will allow the admin (such as the owner and staff admin) to monitor staff availability, track monthly expenses, check feedback, and handle notifications.

1.2 Product Scope

The items within our scope are:

- Online payment
- User registration and profile management
- Real-time tour availability / airport transfer
- Booking/itinerary management for both users and admins
- Customer reviews and rating
- Contact and support chat

The items out of scope are:

- No third-party tour aggregation
- No flight or hotel booking integration in the initial release
- No mobile app (for initial launch)

1.3 Intended Audience and Document Overview

Our target market consists of:

- Tourists planning DIY trips to Japan
- UncleSam Owner and Employees

Early Adopters:

- UncleSam owner and staff

- Tourists that create their own itinerary
- Tourists seeking alternative destinations travel without compromising on premium experiences.

1.4 Definitions, Acronyms and Abbreviations

AI (Artificial Intelligence) – A system capability that simulates human intelligence to generate itineraries, provide chatbot support, analyze data, and offer travel recommendations.

API (Application Programming Interface) – A set of rules that allows the UncleSam System to communicate with external services such as PayMongo and OpenWeatherMap.

Booking – A confirmed reservation made by a client for a tour or service offered by UncleSam Travel and Tours.

Client – A registered end user who can browse offers, manage their account, contact support, and leave reviews, but does not directly manage bookings.

Concurrency – The ability of the system to handle multiple users accessing or performing actions simultaneously without performance degradation.

Itinerary – A structured travel plan generated or modified by the system, often assisted by AI and weather data, based on confirmed bookings.

OpenWeatherMap – A third-party weather forecasting service integrated into the system to provide real-time weather data for itinerary planning.

Owner – The primary administrative user with full access to system features, including account management, bookings, reports, and analytics.

PayMongo – A third-party payment gateway used to securely process online payments and transactions within the system.

RBAC (Role-Based Access Control) – A security mechanism that restricts system functionality and data access based on user roles (Owner, Staff, Client).

SRS (Software Requirements Specification) – A formal document that describes the functional, non-functional, and quality requirements of the UncleSam System.

Staff – Authorized personnel who assist the Owner in managing tours, bookings, and customer support with limited administrative privileges.

Third-Party Integration – The use of external services (e.g., payment gateways, weather APIs, AI services) that operate outside the core system.

TLS (Transport Layer Security) – A cryptographic protocol used to encrypt data transmitted between users and the system to ensure secure communication.

UI (User Interface) – The web-based visual components through which users interact with the UncleSam System.

Uptime – The percentage of time the system remains operational and accessible to users, excluding scheduled maintenance.

1.5 Document Conventions

This Software Requirements Specification (SRS) document follows the **IEEE SRS formatting and documentation standards** to ensure clarity, consistency, and ease of understanding for all readers.

Formatting Conventions

- The document uses Arial font, size 11 or 12, for all body text.
- All text is single-spaced with 1-inch margins on all sides.
- Section and subsection headings follow the numbering format defined in the IEEE SRS template.
- Italicized text is used for comments, notes, and instructional placeholders.
- Bullet points and numbered lists are used where appropriate to improve readability.

Requirement Naming Conventions

- Functional requirements are labeled using unique identifiers (e.g., F1, F2).
- Performance requirements are labeled P1, P2, ….
- Safety and security requirements are labeled S1, S2, ….
- Software quality attributes are labeled Q1, Q2, ….
- These identifiers support traceability across the document.

Terminology and Language Usage

- The word **"shall"** indicates a mandatory requirement.
- The word **"should"** indicates a recommended but non-mandatory requirement.
- The word **"may"** indicates an optional feature.
- User roles such as **Owner**, **Staff**, and **Client** are capitalized to distinguish them from general terms.

Figures and Diagrams

- Figures (e.g., UI screenshots and diagrams) are labeled sequentially (e.g., *Figure 01, Figure 02*).
- UML diagrams follow standard UML notation and are referenced within the text when applicable.

1.6 References and Acknowledgments

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2 Overall Description

2.1 Product Overview

The project is a system that provides the users the flexibility to customize itinerary with the help of AI, weather advisory updates that recommends alternative destinations if the area is affected, and a 24/7 AI chatbot.

2.2 Product Functionality

- Log in
- Booking management
- Itinerary management
- AI chat support
- AI-assisted itinerary
- Weather forecast
- Online payment
- Report and analytics
- Leave and view reviews

2.3 Design and Implementation Constraints

C1. Software Design Methodology Constraint

The system design shall follow the **COMET (Concurrent Object Modeling and Architectural Design Method)** methodology for software development.
This includes the use of object-oriented analysis, concurrent modeling, and architectural decomposition.
Rationale: Required by course standards and ensures structured, maintainable system design.

C2. Modeling Language Constraint

All system models and diagrams shall be created using the **Unified Modeling Language (UML)**, including but not limited to:

- Use Case Diagrams
- Class Diagrams
- Sequence Diagrams
  Rationale: UML is required for standard documentation and design consistency.

C3. Web-Based System Constraint

The UncleSam System shall be implemented as a **web-based application** accessible through modern web browsers.
No native mobile application shall be developed in the initial release.
Rationale: Reduces development cost and aligns with the defined project scope.

C4. Third-Party Integration Constraint

The system shall integrate with the following external services:

- **PayMongo** for payment processing
- **OpenWeatherMap** for real-time weather forecasting
- **AI service API** (e.g., ChatGPT API) for itinerary generation and chatbot support
  Rationale: Core system functionality depends on these services.

C5. Payment Handling Constraint

All financial transactions shall be processed exclusively through **PayMongo**.
The system shall not store or process raw credit or debit card information.
Rationale: Ensures PCI-DSS compliance and reduces security liability.

C6. Security and Privacy Constraint

The system shall comply with the **Data Privacy Act of 2012 (Republic Act No. 10173)**.
All communications shall use **HTTPS/TLS** encryption.
Rationale: Required for legal compliance and user data protection.

C7. Performance and Cost Constraint

AI-powered features (e.g., itinerary generation) shall only be accessible after a confirmed booking.
Rationale: Minimizes operational costs related to AI usage as defined by the business policy.

2.4 Assumptions and Dependencies

Assumptions

- **API Reliability and Availability:** It is assumed that third-party services (like the weather forecast service and the currency conversion service, e.g., OpenWeatherMap, FXRates) provide reliable and available APIs for real-time data integration.
- **PayMongo API:** It is assumed that the PayMongo API documentation is comprehensive and that the service remains operational and accessible for secure payment processing throughout the system's lifecycle.
- **Staff Training and Availability:** It is assumed that the UncleSam staff (including the owner and staff admin) will be adequately trained and available to use the basic admin dashboard to monitor staff availability, track monthly expenses, and handle notifications.

Dependencies

- **AI/Chatbot/Itinerary Component:** The successful implementation of the AI-assisted itinerary generator and 24/7 AI chat support is directly dependent on the availability and correct integration of a chosen Artificial Intelligence component (e.g., ChatGPT API).
- **External Data Sources:** The real-time weather forecast feature is dependent on a stable external weather API (e.g., OpenWeatherMap).
- **Payment Services:** The system is dependent on the proper functioning and security compliance of the PayMongo payment gateway for all online transactions.

TO DO: Provide a short list of some major assumptions that might significantly affect your design.>

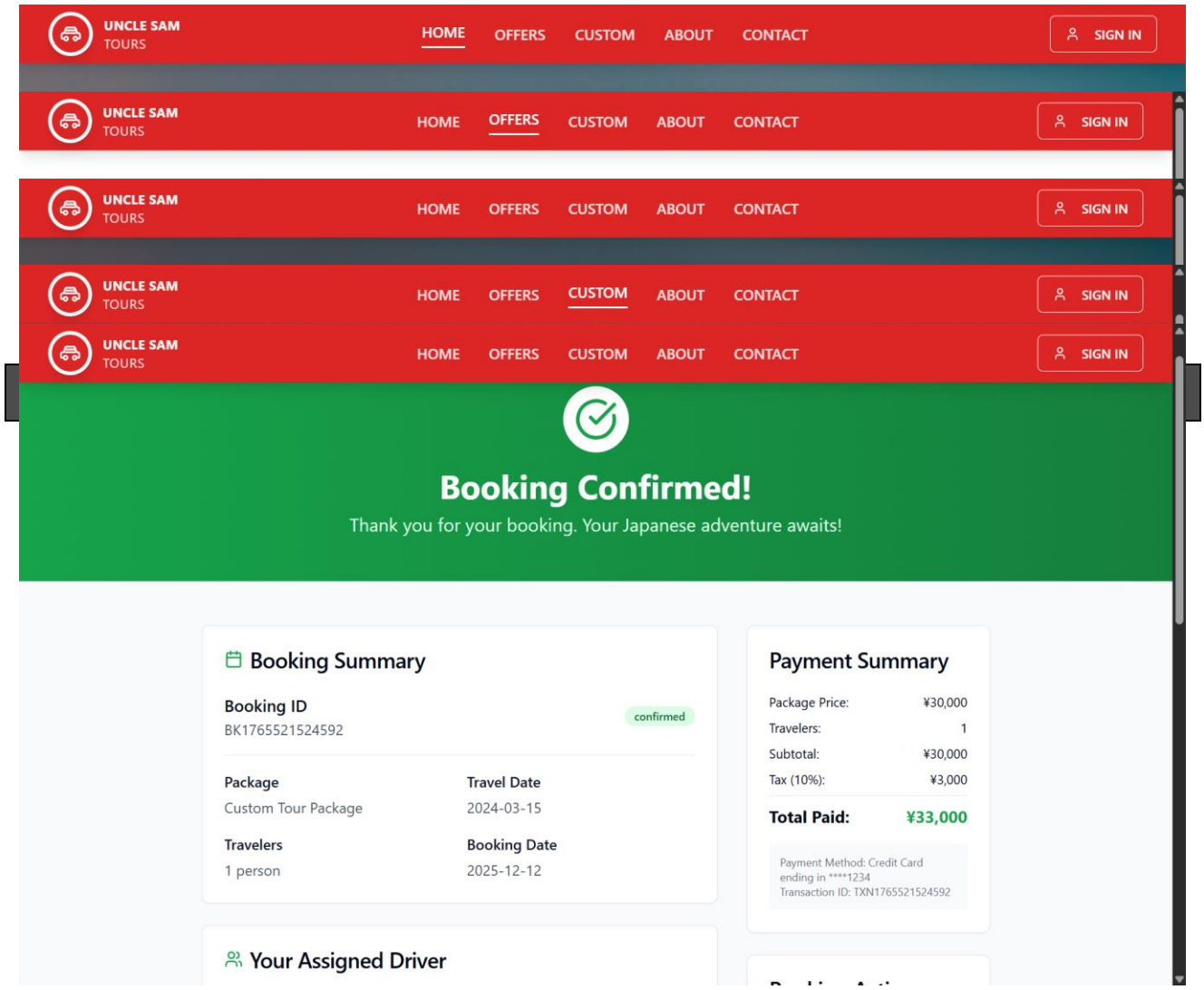


Figure 05. Confirm Boooking

<Describe the logical characteristics of each interface between the software product and the users. For your project, you only need to be concerned with the main thermostat (not the mobile app) and can use the graphic from the project description as the basis for your user interface.

TO DO: Provide the graphic for the thermostat user interface and provide a basic description as to how users will interact (e.g. tough screen, menus, etc.).>

3.1.2 Hardware Interfaces

The UncleSam Travel and Tours system is primarily a web-based application and will not interface directly with specialized hardware devices. However, it requires a reliable environment to support its web-based and administrative functions.

Client Devices:

- **Description:** Standard desktop computers, laptops, tablets, and smartphones used by Clients, Staff, and Owners to access the web application. The front-end shall be responsive to support modern browsers and mobile devices.
- **Interfaces:** Standard internet protocols (e.g., HTTPS/TLS).

3.1.3 Software Interfaces

The UncleSam Travel and Tours system interacts with external services to provide core functionality, and its entire architecture relies on secure Application Programming Interface (API) communication. Below are the detailed descriptions of these connections:

- **Databases:**
  - The primary database is **PostgreSQL** (via **Supabase**). This is used to store all system data, including user accounts, tour listings, booking details, and analytics.
- **Operating Systems:**
  - The system is hosted on cloud platforms (**Vercel** and **Supabase**) which abstract the underlying operating system. The system must be deployable on various cloud environments without code modification.
- **Tools and Libraries:**
  - **Frontend Framework: Next.js** (with **Tailwind CSS**). This builds the user interfaces and dynamic elements for the Client, Staff, and Owner dashboards.
  - **Backend Framework: Nest.js**. This handles server-side logic, manages RESTful API requests and responses, and secures communication with the database and external services.
  - **External Service APIs:**
    - **PayMongo:** Used as the third-party payment gateway for secure transaction processing.

- **OpenWeatherMap:** Used for real-time weather data retrieval for itinerary suggestions.
- **AI Service API (e.g., ChatGPT API):** Used to power the AI-assisted itinerary generation and 24/7 chatbot support.

- **API Protocols:**
  - **RESTful API:** This will define the communication protocols between the frontend, backend, and all external services (PayMongo, OpenWeatherMap, AI Service).
  - **HTTPS/TLS:** Used for all data transmission to ensure secure, encrypted communication.

3.2 Functional Requirements

< Functional requirements capture the intended behavior of the system. This behavior may be expressed as services, tasks or functions the system is required to perform. This section is the direct continuation of section 2.2 where you have specified the general functional requirements. Here, you should list in detail the different product functions.

Authentication and User Management

FR1.1 (Login) The system shall allow Clients, Staff, and Owners to log in by providing a valid email and password.

FR1.2 (Invalid Login Handling) The system shall validate the format of the email and password, and display an "Invalid Credentials!" message upon failed authentication.

FR1.3 (Missing Fields) The system shall prevent submission if required login fields are empty and prompt the user with "Missing Information!".

FR2.1 (View Accounts) The system shall allow the Owner to view a comprehensive list of all existing Staff and Client accounts.

FR2.2 (Create Staff Account) The system shall allow the Owner to create new Staff accounts, capturing required personal details and assigning specific roles/permissions.

FR2.3 (Update Staff Account) The system shall allow the Owner to modify existing Staff account details, roles, and permissions.

FR2.4 (Delete Staff Account) The system shall allow the Owner to securely delete Staff accounts.

Tour and Offer Management
FR3.1 (Add Tour) The system shall allow the Staff and Owner to add new tour listings, including input fields for tour details, itinerary, pricing, and availability.

FR3.2 (Edit Tour) The system shall allow the Staff and Owner to modify the details of any existing tour listing.

FR3.3 (Delete Tour) The system shall allow the Staff and Owner to delete existing tour listings.

FR3.4 (Tour Search Interface) The system shall provide a public-facing search interface allowing Clients to search for tours using criteria such as destination, date, and itinerary keywords.

FR3.5 (Display Search Results) The system shall retrieve and display all tour listings that match the Client's specified search criteria.

Itinerary and Booking Management
FR4.1 (Client Booking View) The system shall allow the Client to view a personalized list of their current and past bookings and itineraries.

FR4.2 (Client Modify/Cancel) The system shall allow the Client to modify or cancel their existing bookings, subject to business rules.

FR4.3 (Staff/Owner Booking View) The system shall allow the Staff and Owner to view all Client booking details and statuses.

FR4.4 (Staff/Owner Booking Action) The system shall allow the Staff and Owner to manually confirm or cancel a Client's booking.

FR4.5 (Real-time Status Update) The system shall update the booking status in real-time following any modification, confirmation, or cancellation action.

FR4.6 (Notification on Update) The system shall automatically send a notification (via email or in-app) to the Client upon any update to their booking status.

AI and Support Services
FR5.1 (AI Itinerary Modification) The system shall integrate an AI chatbot that can modify the details of a Client's booked itinerary based on natural language input, after Client confirmation.

FR5.2 (AI Weather Suggestion) The system's AI component shall integrate with a weather forecast service to suggest alternative destinations or itinerary changes due to adverse weather.

FR6.1 (Contact Support Access) The system shall provide an accessible support channel (AI chatbot) where Clients can send inquiries regarding their trip.

FR6.2 (Staff Escalation) The system shall automatically route Client inquiries to the appropriate Staff member if the AI chatbot cannot provide a resolution.

FR6.3 (Support Ticket Generation) The system shall automatically generate a formal support ticket and escalate the issue to the Owner/Staff if the initial inquiry remains unresolved.

Reporting and Analytics
FR7.1 (Analytics Dashboard) The system shall provide an exclusive dashboard for the Owner to access reports and analytics.

FR7.2 (Report Generation) The system shall generate reports on key business metrics, including booking trends, revenue, and customer feedback summaries.

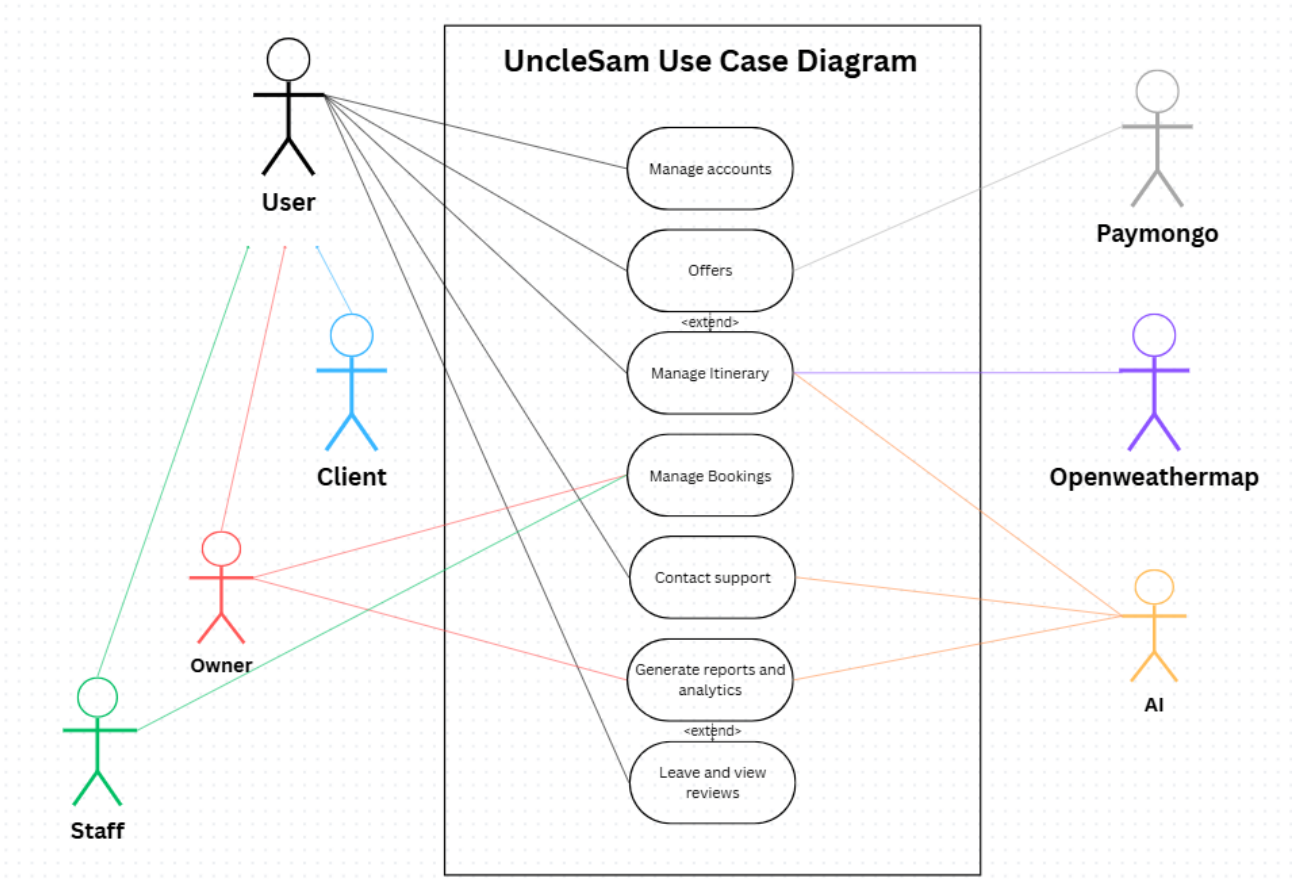
FR7.3 (Data Visualization) The system shall display reporting data using visual charts and detailed summaries to aid Owner decision-making.

FR7.4 (Export Reports) The system shall allow the Owner to export generated reports in standard formats (e.g., PDF, CSV).

Reviews and Feedback
FR8.1 (Leave Review) The system shall allow a Client to submit a rating and comment (review) only for a tour that has been marked as completed.

FR8.2 (Display Review) The system shall store and display submitted reviews and ratings on the respective tour listing page, visible to all users.…

3.3 Use Case Model



3.3.1 Use Case #1 (use case name and unique identifier – e.g. U1)

TO DO: Provide a specification for each use case diagram

Use Case Name: Login

Use Case ID:	*UC-01*
Author	*Abrech Dela Cruz*

Purpose	*Allow users/staff/owner to login accounts on the website.*
Requirement Traceability	*BR-01*
Priority	*High*
Preconditions	• *System must be online*
Postconditions	• *User logged in an account* • *Redirected to Login page*
Actors	*Client, Staff, Owner*
Flow of Actions	
	Basic Flow 1. *Client navigates to the login page.* 2. *Client enters required information (name, email, and password).* 3. *Client clicks login.* ***Alternative Flow: Invalid Email Address*** 1. *Client navigates to the login page.* 2. *Client enters required information (name, email, and password).* 3. *Click submit button.* 4. *The user is prompted with a message "Invalid Credentials!"* 5. *The page stays in the login page.* ***Missing Required Fields*** 1. *Client navigates to the registration page.* 2. *Client enters some information (name, email, and password).* 3. *Click submit button.* 4. *The client is prompted with a message "Missing Information!"* 5. *The page stays in the registration page.*

Use Case Name: Manage Accounts

Use Case ID:	*UC-02*
Author	*Jesse Isaiah G. Sunga*
Purpose	*Enable owner to efficiently manage staff accounts, including creating, updating, delete accounts to maintain secure and organized staff operations*
Requirement Traceability	*BR-02*
Priority	*High*
Preconditions	• *Owner is logged in* • *System is online*
Postconditions	• *Staff accounts are created, updated or deleted as requested* • *Staff roles and permissions are properly assigned and updated*

	• *Owner receives confirmation of changes* • *Owner can view client's account*
Actors	*Owner*
Extends	
Flow of Actions	***Basic Flow*** 1. *Owner logs into the system* 2. *Navigates to the "Account Management" section* 3. *Views the list of existing staff/client accounts* 4. *Creates a new staff account by entering required details and assigning roles/ permissions* 5. *Updates existing staff account information or modifies roles/ permission as needed* 6. *Deletes staff accounts when necessary*

Use Case Name: Offers

Use Case ID:	*UC-03*
Author	*Abrech Dela Cruz*
Purpose	*Allow Owner and Staff to manage tour listings* *Allow users to search tours on the website*
Requirement Traceability	*BR-03*
Priority	*High*
Preconditions	• *User must be logged in* • *System must be online*
Postconditions	• *User must be able to search for tour listings*
Actors	*Owner, Staff*
Flow of Actions	
	Basic Flow: Viewing tour listings 1. *Client navigates to search interface.* 2. *Client enters search criteria (destination, date, itinerary).* 3. *Client submits search.* 4. *System retrieves matching tours and displays results.* ***Alternative Flow: Client booking a tour*** 1. *Client navigates to a tour listing.* 2. *Client chooses a date, number of pax, payment type (down payment or full payment).* 3. *Client makes payment.* 4. *Client receives a booking confirmation.* ***Alternative Flow: Client booking multiple tours*** 5. *Client navigates to custom tourss.* 6. *Client chooses a date, number of pax, payment type (down*

	payment or full payment). *7. Client makes payment.* *8. Client receives a booking confirmation.* *9. Client can choose a location, date, number of pax again.* ***Alternative Flow: Adding tour listings*** 1. *Staff/Owner logs in to the staff or owner dashboard.* 2. *Staff/Owner navigates to the "Manage Tours" section.* 3. *Staff/Owner chooses to add a new tour, edit an existing tour, or delete a tour.* 4. *For adding or editing, Staff/Owner enters required tour details.* 5. *The system validates input and saves changes.* 6. *System confirms success and updates tour listing on the client-facing website.*

Use Case Name: Manage Itinerary

Use Case ID:	*UC-04*
Author	*Shanteiy Camama*
Purpose	*Allow Customer/Staff/Owner to manage itinerary*
Requirement Traceability	*BR-04*
Priority	*High*
Preconditions	• *User/ Staff/ Owner is logged in* • *System online* • *There are existing bookings associated with the logged in user or managed by staff/ owner*
Postconditions	• *Bookings are displayed accurately* • *Users/Staff/Owner can view, update, modify, or cancel bookings as permitted* • *System reflects booking status changes in real-time* • *Notifications are sent upon booking updates or cancellations*
Actors	*Client, Owner, Staff, AI*
Extends	*UC-05*
Flow of Actions	***Basic Flow*** 1. *User logs into the system* 2. *Navigate to the "Bookings" section* 3. *System displays a list of current and past bookings relevant to the client* 5. *Select a booking to view details* 6. *Choose to modify, cancel, or update the booking if allowed* 7. *Confirm any changes; system processes updates and sends notifications*

	Alternative Flow: AI-assisted itinerary 1. *User inputs a prompt in AI chatbot in itinerary management* 2. *AI will modify the itinerary in detail depending on client's input* 3. *AI will suggest an alternative destination depending on weather forecast* 4. *AI will proceed to modify the itinerary as permitted by the client*

Use Case Name: Manage and Cancel Bookings

Use Case ID:	*UC-05*
Author	*Clarice V. Benoman*
Purpose	*Allow Staff/Owner to view and cancel bookings*
Requirement Traceability	*BR-05*
Priority	*High*
Preconditions	• *Staff/Owner is logged in.* • *System is online.*
Postconditions	• *Staff/Owner receives the booking confirmation and details*
Actors	*Staff/Owner*
Extends	*UC-06*
Flow of Actions	***Basic Flow*** 1. *Go to Staff/Owner dashboard* 2. *View client booking* 3. *Manage booking if it should be confirmed or cancelled* 4. *System sends inquiry to support staff and client* 5. *Client receives notification message or email*

Use Case Name: Contact Support

Use Case ID:	*UC-06*
Author	*Clarice V. Benoman*
Purpose	*Allow Clients to contact support service.*
Requirement Traceability	*BR-06*
Priority	*High*
Preconditions	• *Client is logged-in* • *System is online* • *Client has concerns regarding the trip* • *Client needs immediate inquiry response/s*
Postconditions	• *Inquiry is submitted*

	• *Inquiry is received and reviewed*
Actors	*Client, Owner, Staff, AI*
Flow of Actions	***Basic Flow*** 1. *Login* to account 2. *Press AI chatbot button* 3. *Send inquiry* 4. *System sends inquiry to staff (upon availability) and is also submitted to the AI system* 5. *Wait for response* 6. *User receives response accordingly* ***Alternate Flow*** 1. *If issue is unresolved, support ticket will be submitted* 2. *Client will be directed to the Owner/Staff* 3. *Wait for response* 4. *User receives response accordingly*

Use Case Name: Generate Reports and Analytics

Use Case ID:	*UC-07*
Author	*Shanteiy A. Camama*
Purpose	*Enable the owner to view detailed reports and analytics to monitor tour performance, bookings and business metrics for informed decision-making*
Requirement Traceability	*BR-07*
Priority	*High*
Preconditions	• *Owner is logged-in* • *System is online*
Postconditions	• *Owner successfully accesses up to date reports and analytics dashboard* • *Owner can generate and export reports as needed* • *Owner uses insights to improve business operations*
Actors	*Owner*
Extends	
Flow of Actions	***Basic Flow*** 1. *Owner logs into the system* 2. *Navigates to the "Analytics" section* 3. *Selects the type of report or analytics dashboard to view; booking trends, revenue, customer feedback* 4. *System displays the requested report with visual charts and data summaries* 5. *Owner reviews the data and exports reports if needed*

Use Case Name: Leave and View Reviews

Use Case ID:	*UC-08*
Author	*Abrech Dela Cruz*
Purpose	*Allow Clients to leave and view reviews*
Requirement Traceability	*BR-08*
Priority	*High*
Preconditions	• *User is logged-in* • *System is online* • *User has completed a tour*
Postconditions	• *Review is published*
Actors	*Client, Owner, Staff*
Extends	
Flow of Actions	***Basic Flow*** 1. *Login to account* 2. *Navigate to "My Bookings"* 3. *Click "Leave a Review" on a completed tour* 4. *Enter rating and comment* 5. *Submit the review* 1. *System displays and stores the review*

3.3.2 Use Case #2

…

4 Other Non-functional Requirements

4.1 Performance Requirements

P1. Offer and Itinerary Data Loading
The system shall load offers and itinerary-related data within **3–5 seconds** after user request.
Rationale: Ensures a responsive browsing experience while handling API and database queries.

P2. Booking and Payment Processing Time
Payment transactions via **PayMongo** shall be completed or fail gracefully within **10 seconds**.
Rationale: Avoids duplicate charges and ensures transaction clarity for clients.

P3. AI Itinerary Generation Performance
AI-generated itinerary recommendations shall be produced within **8 seconds** after a confirmed booking.
Rationale: Balances system cost control (AI usage) with acceptable user waiting time.

P4. Policy-Based Access Control Enforcement
The system shall immediately restrict access to AI itinerary features if no booking is detected (**real-time check < 1 second**).
Rationale: Enforces the business policy of booking-first to minimize unnecessary AI computation costs.

P5. Weather Data Retrieval
Weather information from **OpenWeatherMap** shall be retrieved and displayed within **5 seconds**.
Rationale: Ensures itinerary accuracy without significantly delaying the user workflow.

P6. Report and Analytics Generation (Owner Only)
Reports and analytics shall be generated within **10–15 seconds**, depending on data volume.
Rationale: Reports are non-real-time features and can tolerate longer processing times.

P7. Concurrent User Support
The system shall support at least **50 concurrent users** without noticeable performance degradation.
Rationale: Supports peak usage during promotions or travel seasons.

P8. System Availability
The system shall maintain **99% uptime**, excluding scheduled maintenance.
Rationale: Ensures reliability for bookings and client trust.

4.2 Safety and Security Requirements

S1. User Authentication and Identity Verification

The system shall require secure user authentication (username and password) for all user roles (Owner, Staff, Client).
Rationale: Prevents unauthorized access to accounts and system features.

S2. Role-Based Access Control (RBAC)

The system shall enforce role-based access control to restrict access based on user roles:

- Owners have full system access
- Staff have limited administrative access
- Clients have service-level access only
  Rationale: Prevents accidental or malicious access to sensitive data and features.

S3. Secure Mobile and Web Communication

All data transmitted between the client device (web or mobile) and the system server shall use encrypted connections (HTTPS/TLS).
Rationale: Protects against man-in-the-middle attacks and data interception, especially on public networks.

S4. Payment Security Compliance

All payment transactions shall be handled exclusively through PayMongo, which complies with PCI-DSS standards.
The system shall not store credit or debit card details.
Rationale: Reduces financial risk and ensures compliance with industry security standards.

S5. Data Privacy and Personal Information Protection

The system shall protect all personal user data in compliance with the Data Privacy Act of 2012 (Republic Act No. 10173).
Users shall be informed about data collection and usage policies.
Rationale: Ensures ethical handling of personal information and legal compliance.

S6. Account Lockout and Session Security

The system shall temporarily lock an account after multiple failed login attempts (e.g., 5 attempts).
User sessions shall automatically expire after a period of inactivity.
Rationale: Prevents brute-force attacks and unauthorized session access.

S7. Secure AI and API Access

Access to AI services and third-party APIs (OpenWeatherMap, PayMongo) shall require secured API keys stored on the server side only.
Rationale: Prevents misuse, unauthorized calls, and data leaks.

S8. Logging and Monitoring

The system shall log critical activities such as login attempts, payment events, and administrative actions.
Only the Owner shall have access to system logs.
Rationale: Enables auditing, issue tracing, and incident response.

4.3 Software Quality Attributes

4.3.1 Reliability

Q1. The system shall maintain 99% operational availability, excluding scheduled maintenance.
Q2. The system shall recover gracefully from third-party API failures by displaying fallback messages within 3 seconds.
Q3. Failed transactions shall not result in duplicate bookings or charges.

Design Approach:

Use error handling and retry mechanisms for PayMongo and OpenWeatherMap APIs

Implement database transaction rollbacks for booking and payment operations

4.3.2 Usability

Q4. New clients shall be able to complete a booking within **3 minutes** without external assistance.
Q5. The system shall provide clear UI feedback for restricted features (e.g., itinerary access before booking).

Design Approach:

- Role-based UI components (Owner, Staff, Client)
- Clear system prompts explaining booking-first policy

4.3.3 Maintainability

Q6. *The system shall use a* ***modular architecture*** *separating business logic, AI services, and third-party integrations.*
Q7. *Changes to one module shall not require modifications to unrelated modules.*
Design Approach:
- *Service-oriented backend design*
- *Centralized configuration for API keys and policies*

4.3.4 Adaptability (Design for Change)

Q8. *The system shall support replacement or addition of third-party APIs (e.g., weather or payment providers) with minimal code changes.*
Q9. *AI features shall be configurable to enable, disable, or limit usage without redeploying the system.*
Design Approach:
- *API abstraction layers*
- *Feature toggles for AI services*

4.3.5 Portability

Q10. *The system shall be deployable on cloud platforms without code modification.*
Q11. *The frontend shall support modern browsers and mobile devices.*
Design Approach:

- *Cloud-ready backend architecture*
- *Responsive web design*

5 Other Requirements

*<This section is **Optional**. Define any other requirements not covered elsewhere in the SRS. This might include database requirements, internationalization requirements, legal requirements, reuse objectives for the project, and so on. Add any new sections that are pertinent to the project.>*

Appendix A – Data Dictionary/ERD

<Data dictionary is used to track all the different variables, states and functional requirements that you described in your document. Make sure to include the complete list of all constants, state variables (and their possible states), inputs and outputs in a table. In the table, include the description of these items as well as all related operations and requirements.>

Appendix B - Group Log

<Please include here all the minutes from your group meetings, your group activities, and any other relevant information that will assist in determining the effort put forth to produce this document>

Appendix C – Test Plan/Test Cases